

Evaluating Limits

The Limit of a Function:

Calculating Limits Graphically:

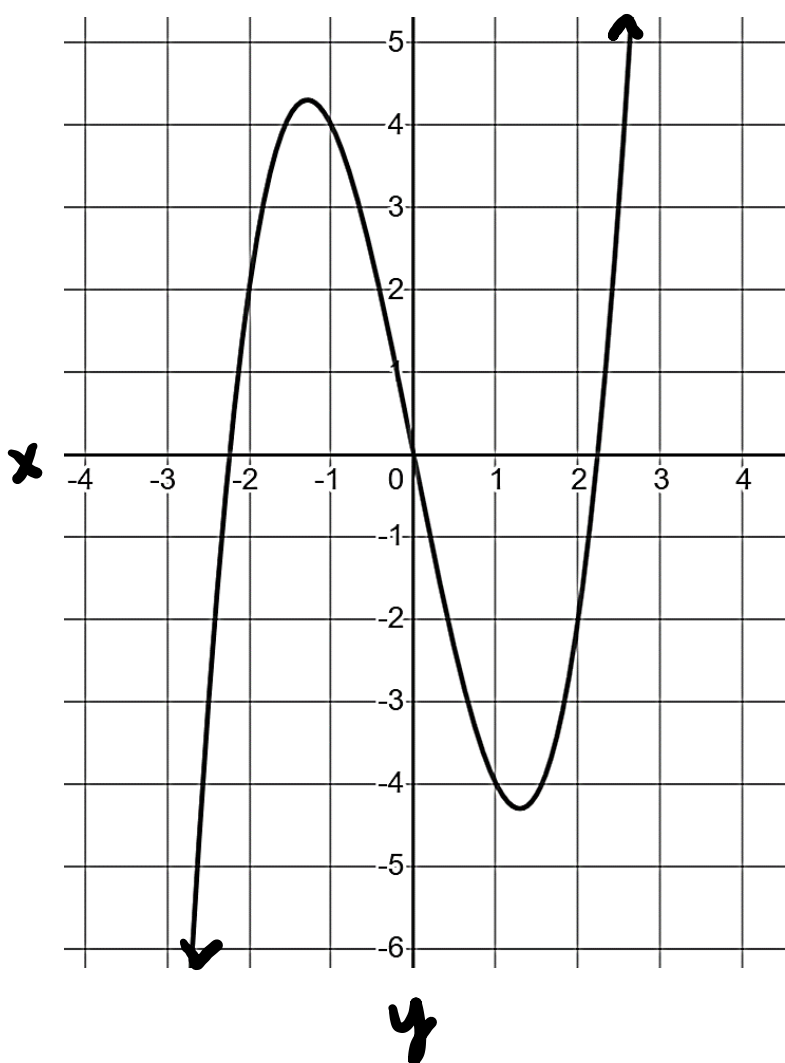
Ex 1: Find the following limits:

$$\lim_{x \rightarrow 0} f(x)$$

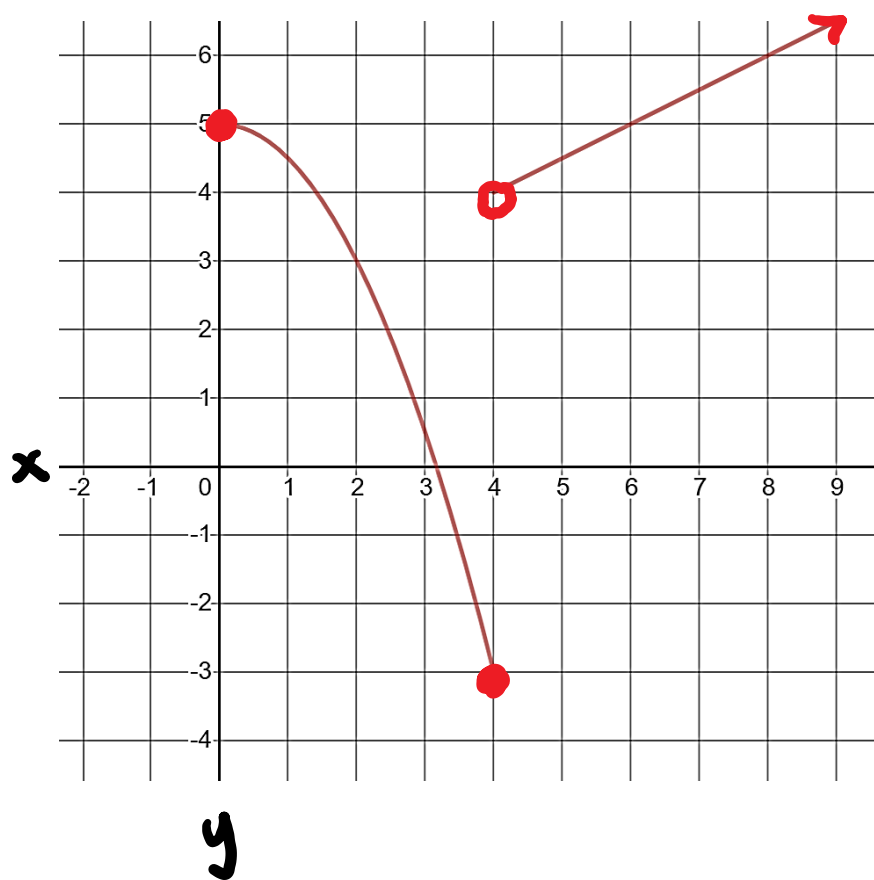
$$\lim_{x \rightarrow -1} f(x)$$

$$\lim_{x \rightarrow 0.5} f(x)$$

$$\lim_{x \rightarrow 2} f(x)$$



One Sided Limits:



Ex 2: Find the following limits:

$$\lim_{x \rightarrow 0^+} f(x)$$

$$\lim_{x \rightarrow 2^-} f(x)$$

$$\lim_{x \rightarrow 4^-} f(x)$$

$$\lim_{x \rightarrow 4^+} f(x)$$

$$\lim_{x \rightarrow 4} f(x)$$

Evaluating Limits Algebraically

Ex 1: Evaluate the limit.

$$\lim_{x \rightarrow 5} \frac{x^2 + 1}{5 + x}$$

Ex 2: Evaluate the limit.

$$\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x^2 - 9}$$

Ex 3: Evaluate the limit:

$$\lim_{x \rightarrow 5} \frac{\sqrt{x-1} - 2}{x-5}$$